

TODO title about Y-haplotype effect on larval transcriptome

Milena Trabert¹, Bianca Sarcani¹, Philipp Kaufmann², and Elina Immonen¹

¹Department of Ecology and Genetics (Evolutionary Biology program), Evolutionary
Biology Centre, Norbyvägen 18D, 75236, Uppsala University, Uppsala

²Philipp's current affiliation

Sample numbers

- Day 14
 - Small-Y: 10 males, 3 females
 - Large-Y: 9 males, 6 females
- Day 16
 - Small-Y: 11 males, 3 females
 - Large-Y: 7 males, 6 females
- Day 18
 - Small-Y: 7 males, 5 females
 - Large-Y: 12 males, 3 females

Additional results

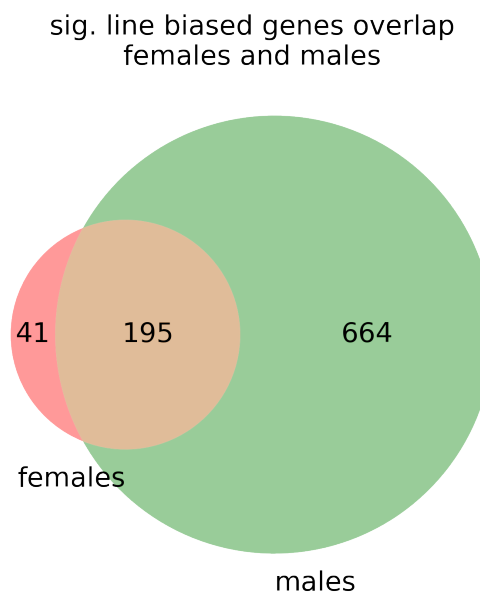


Figure S1: **Line-bias in male and female samples.** The contrast of small-Y - large-Y was tested in male and female samples separately.

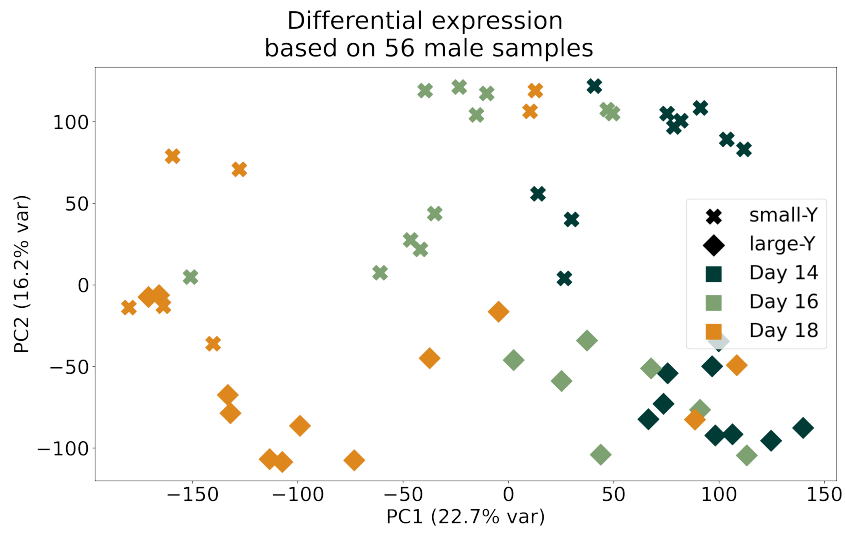


Figure S2: PCA plots of male samples.

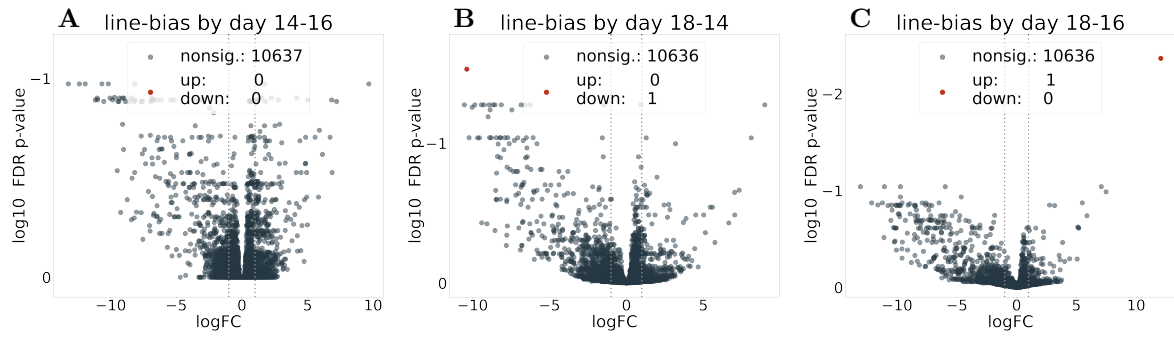


Figure S3: Volcano plot of line-by-day interaction in only male samples. The interaction is calculated between the contrasts of smallY-largeY and each pairwise day comparison, day 14-16 (A), day 18-14 (B) and day 18-16 (C).

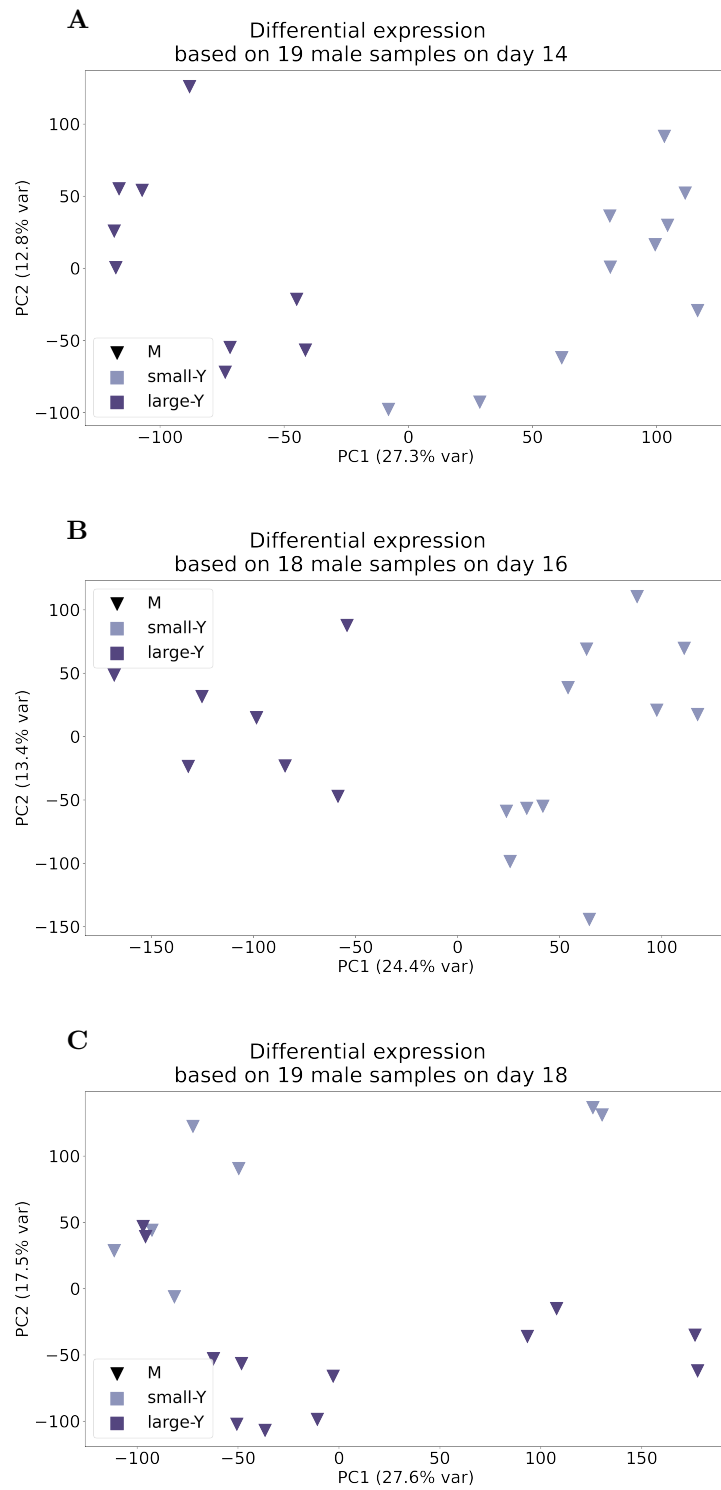


Figure S4: **PCA plots of samples from each time point separately.**

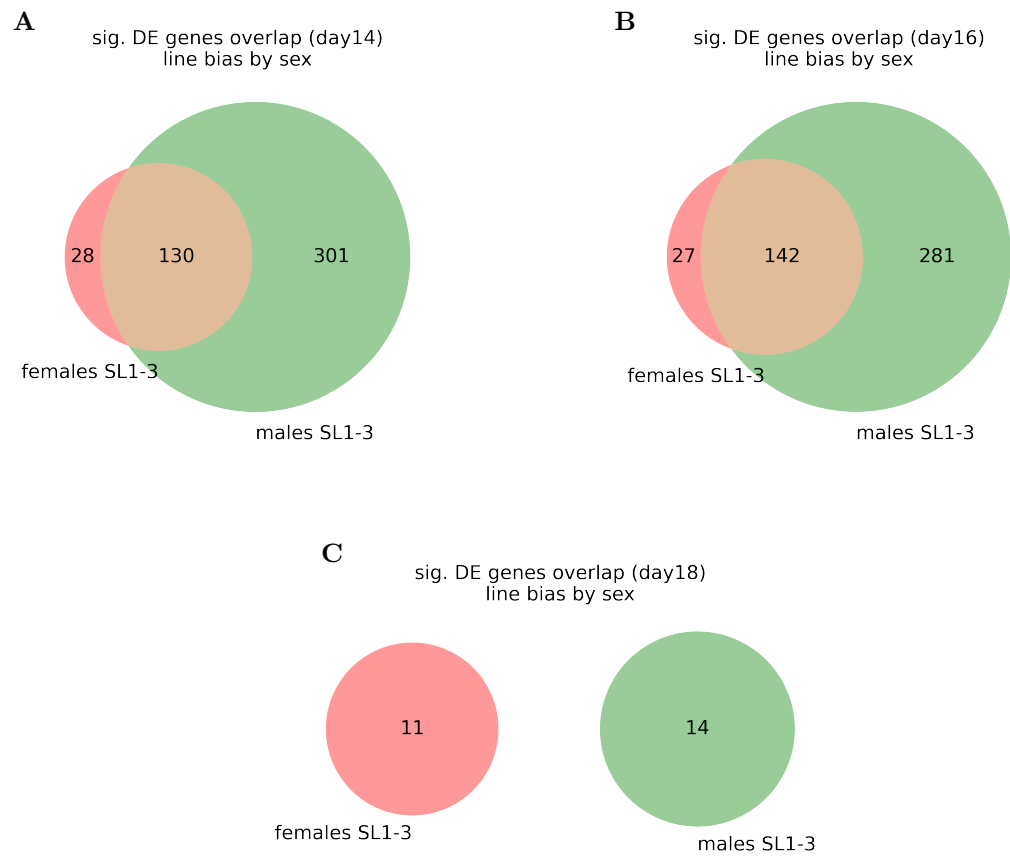


Figure S5: **Overlap of line-biased genes between the sexes.** One overlap for each day, significance calculated on day-separated samples. SL1 refers to the small-Y haplotype and SL3 to the large-Y haplotype.



Figure S6: GO-term enrichment and semantic clustering of male line-biased samples on day 14.

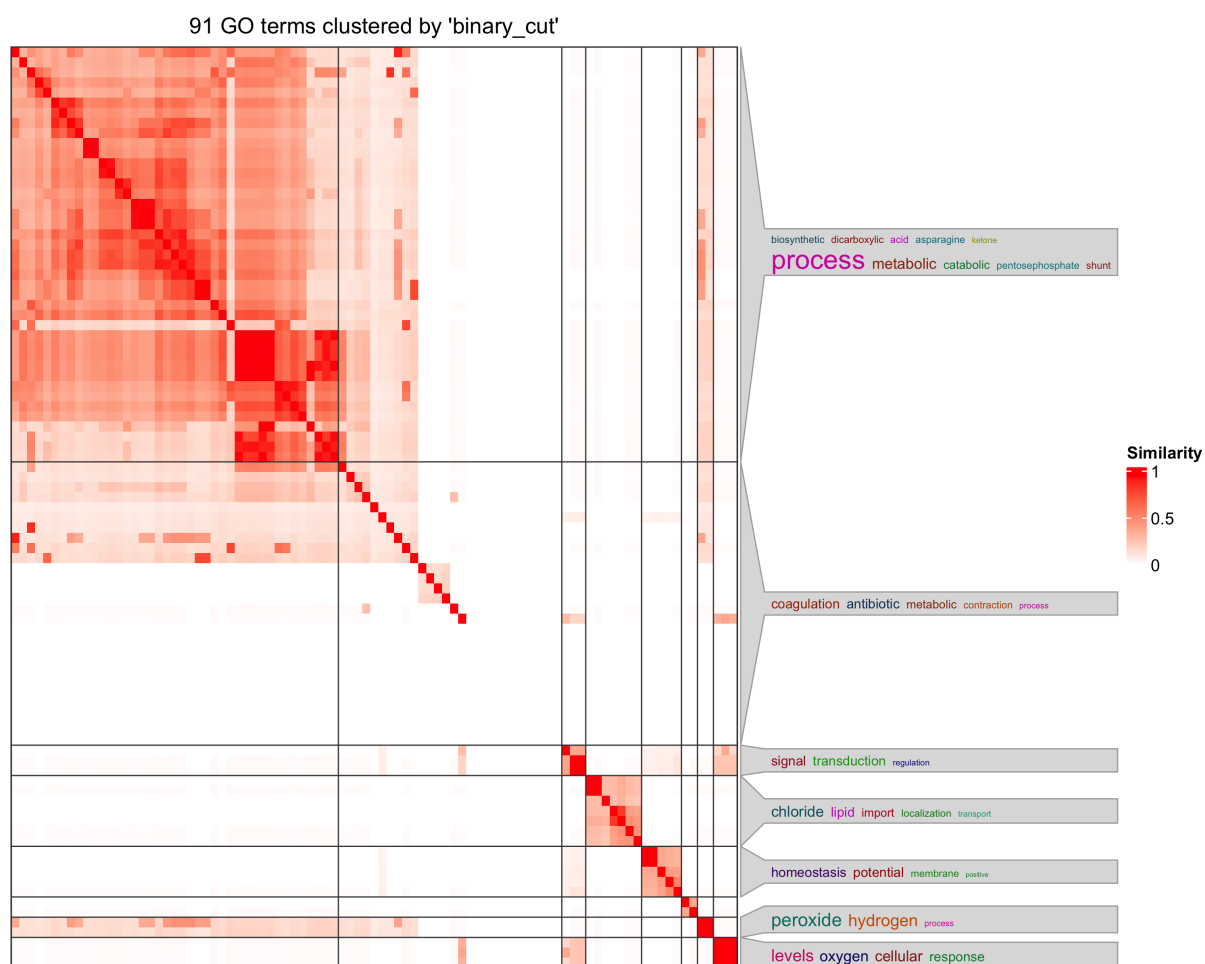


Figure S7: GO-term enrichment and semantic clustering of male line-biased samples on day 16.



Figure S8: **GO-term enrichment and semantic clustering of male line-biased samples on day 18.**

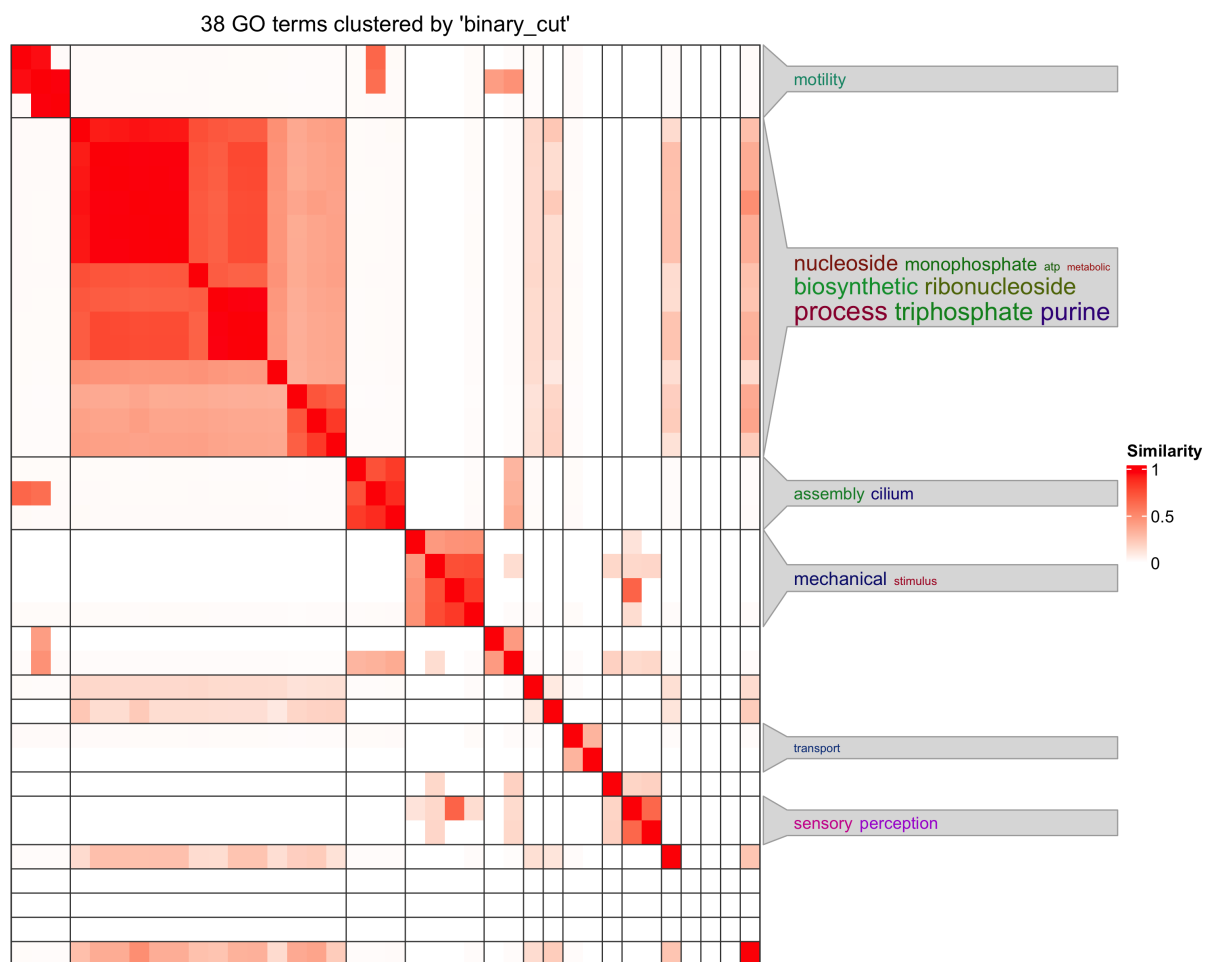


Figure S9: **GO-term enrichment and semantic clustering of genes that are sex-biased.** Full dataset with line as fixed effect, GO terms that are enriched in sex-biased genes of all three time points, intersection as specified in Fig. 2D

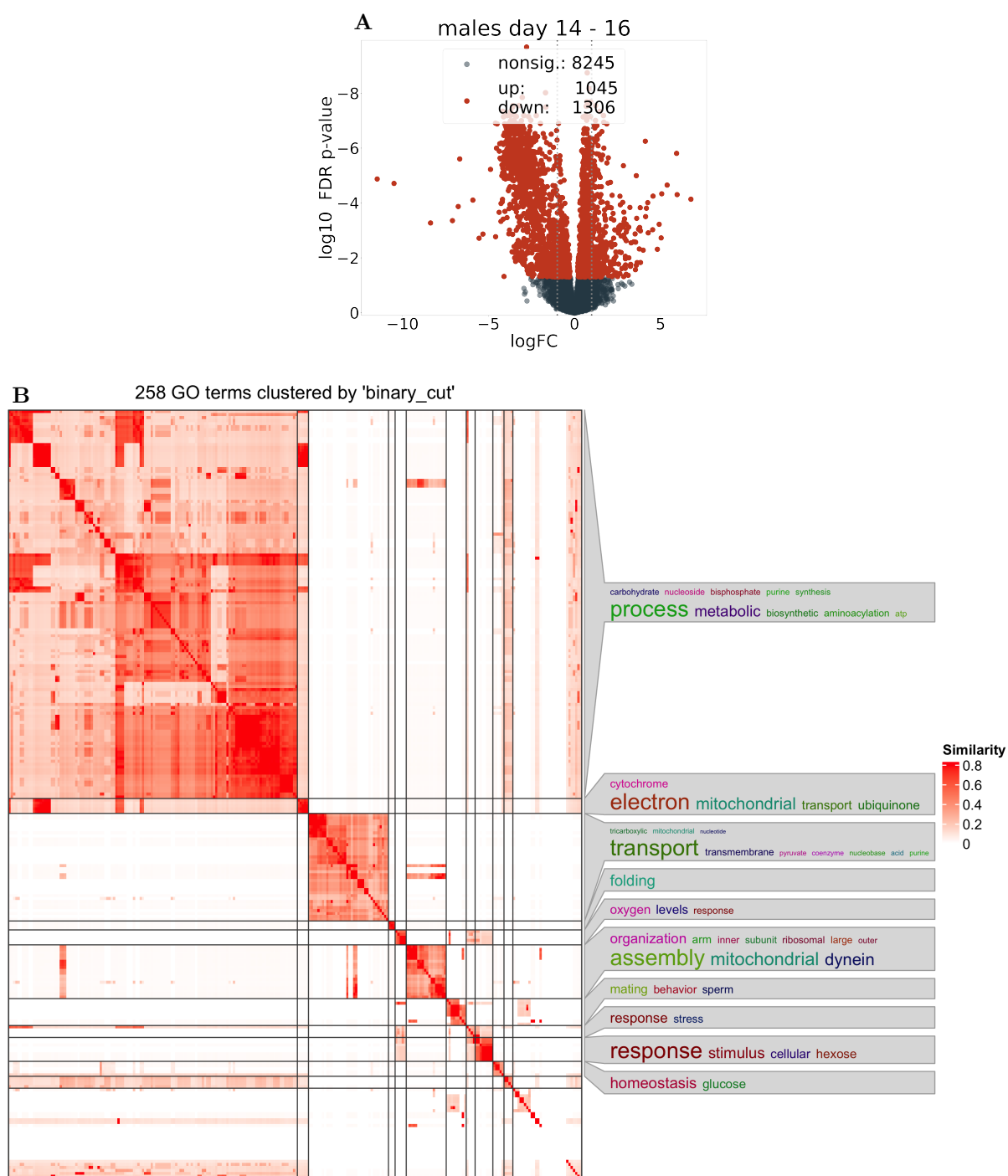


Figure S10: **volcano plot, GO-term enrichment and semantic clustering of genes that are day-biased.** Only male samples with day 14-16 comparison (Fig. 2E intersection). (A) shows the volcano plot, and (B) the semantic clustering of GO-terms that are enriched in sig. DE genes.

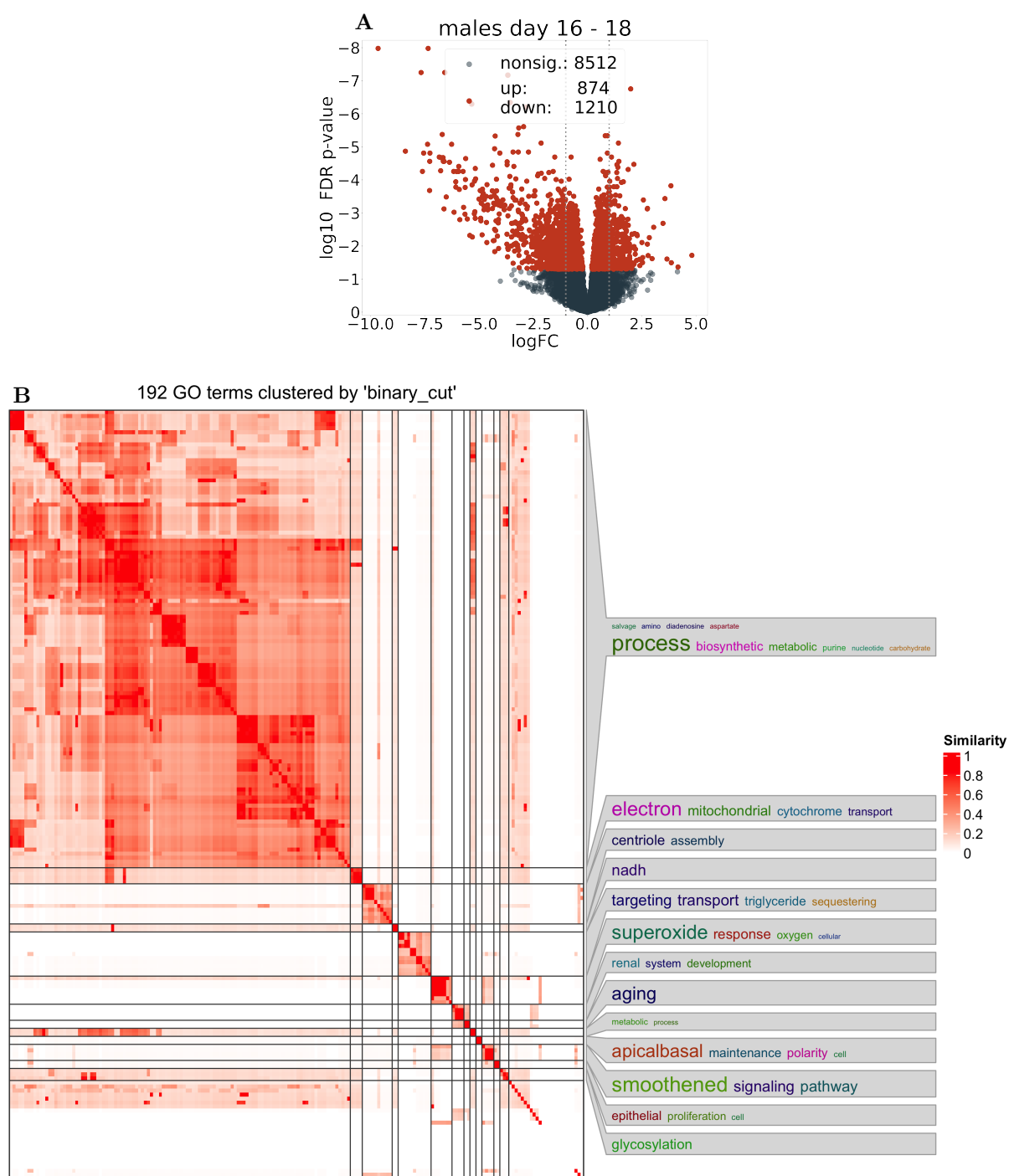


Figure S11: **volcano plot, GO-term enrichment and semantic clustering of genes that are day-biased.** Only male samples with day 16-18 comparison (Fig. 2E intersection) (A) shows the volcano plot, and (B) the semantic clustering of GO-terms that are enriched in sig. DE genes.

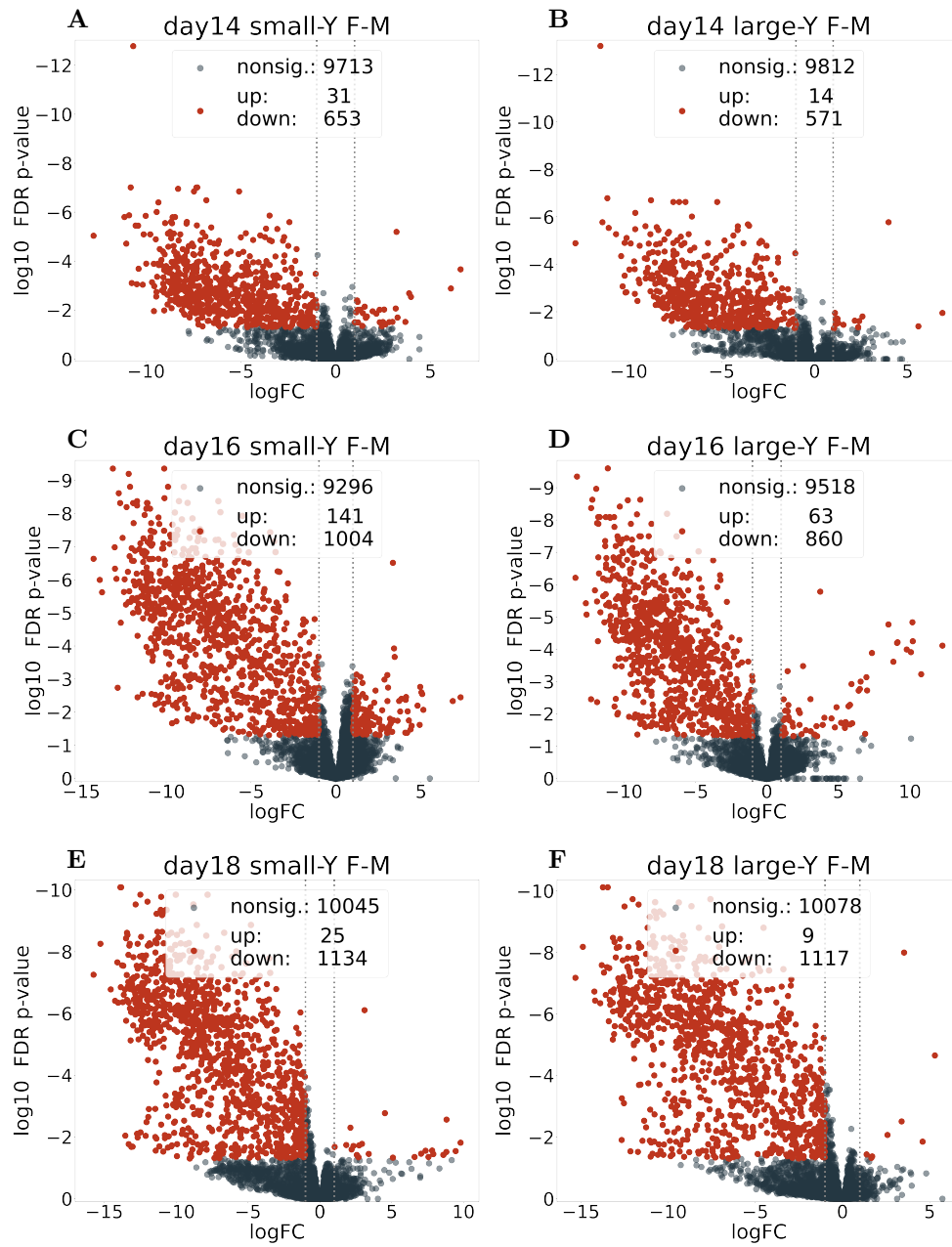


Figure S12: Volcano plot of sex-bias within each line for samples of each time point.

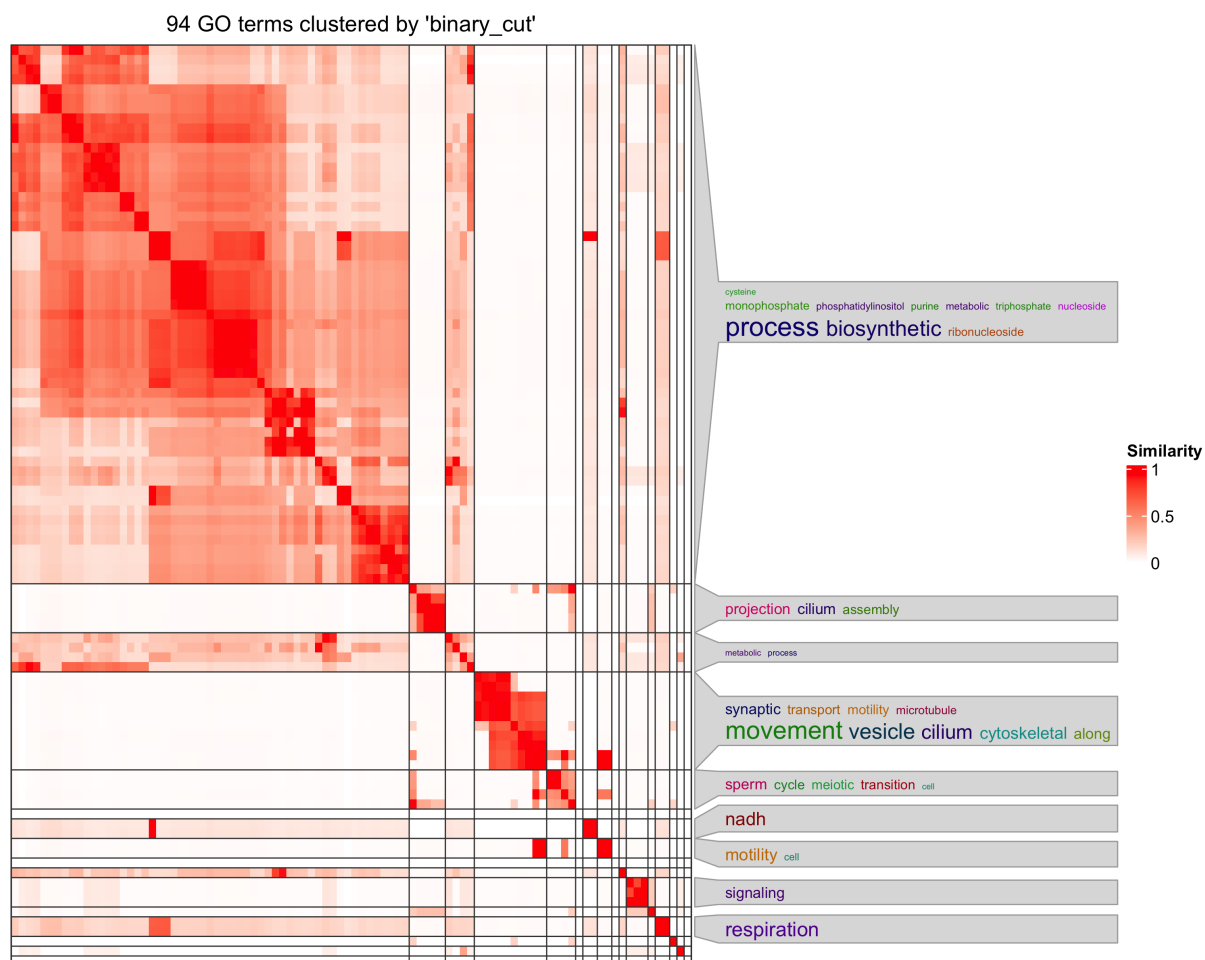


Figure S13: **GO-term enrichment and semantic clustering of genes that are sex-biased on all days.** Sex-bias is estimated separately for each time point in the day-split data, genes that are sex-biased in both lines and all days are determined and GO-enrichment and semantic clustering is performed.

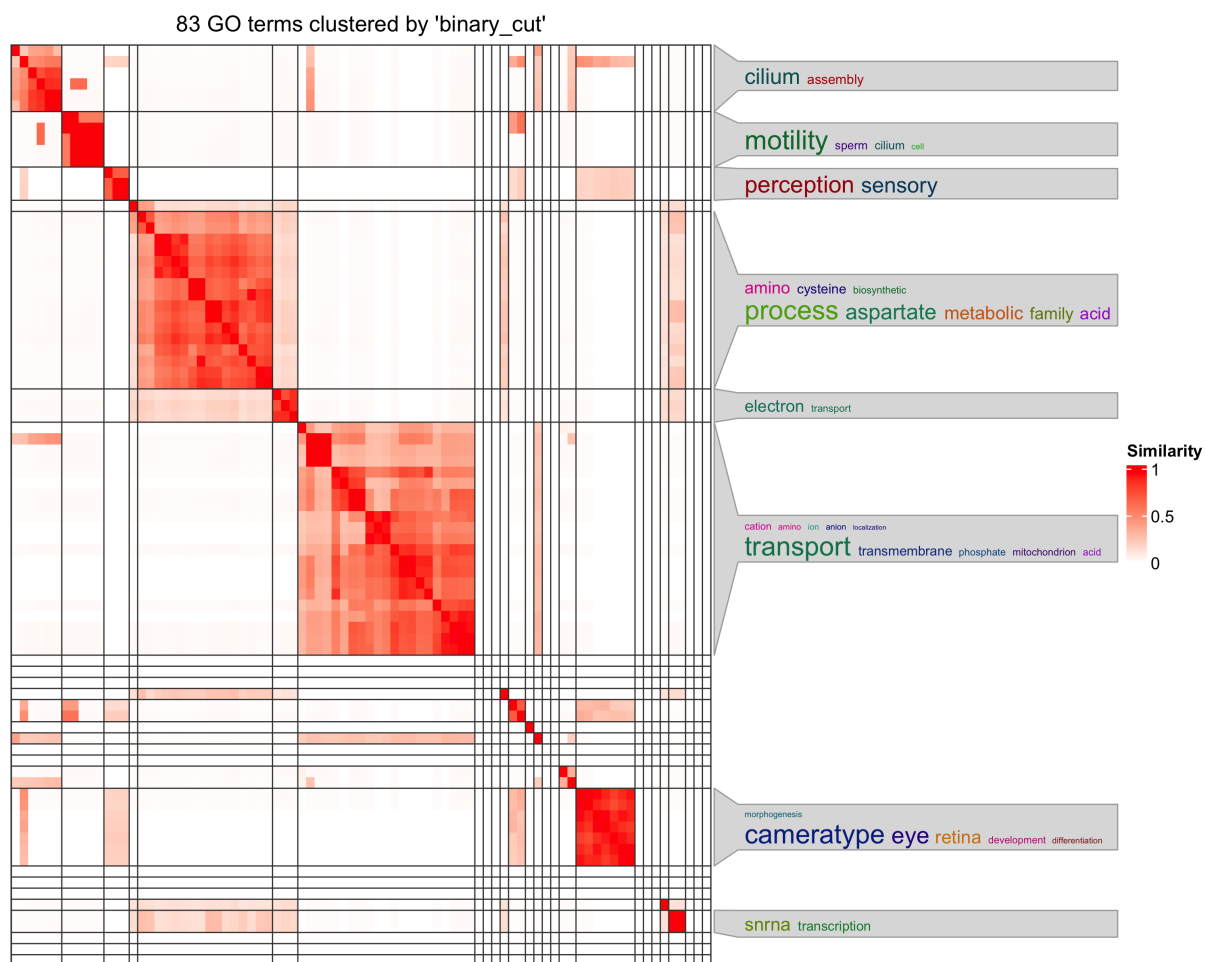


Figure S14: **GO-term enrichment and semantic clustering of genes that are sex-biased on all days and lines except small-Y on day 14.** Sex-bias is estimated separately for each time point in the day-split data, genes that are sex-biased in all conditions except in small-Y lines on day 14 are determined and GO-enrichment and semantic clustering is performed.

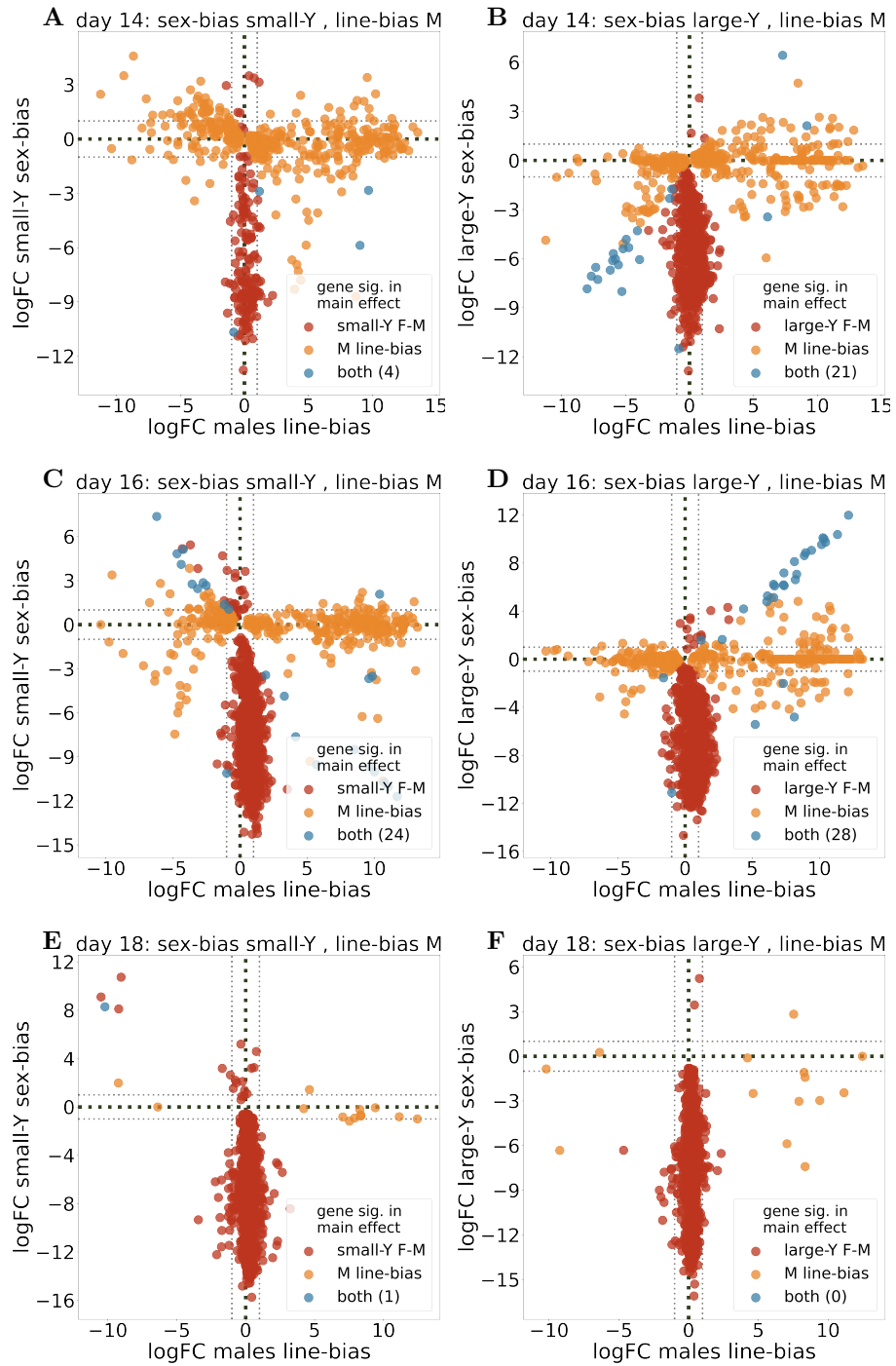


Figure S15: **Intersection of sex bias and line bias across developmental stages day 18.** Log₂FC of sex-bias (females - males contrast) and line-bias (large-Y - small-Y) of day 14, 16, and 18.

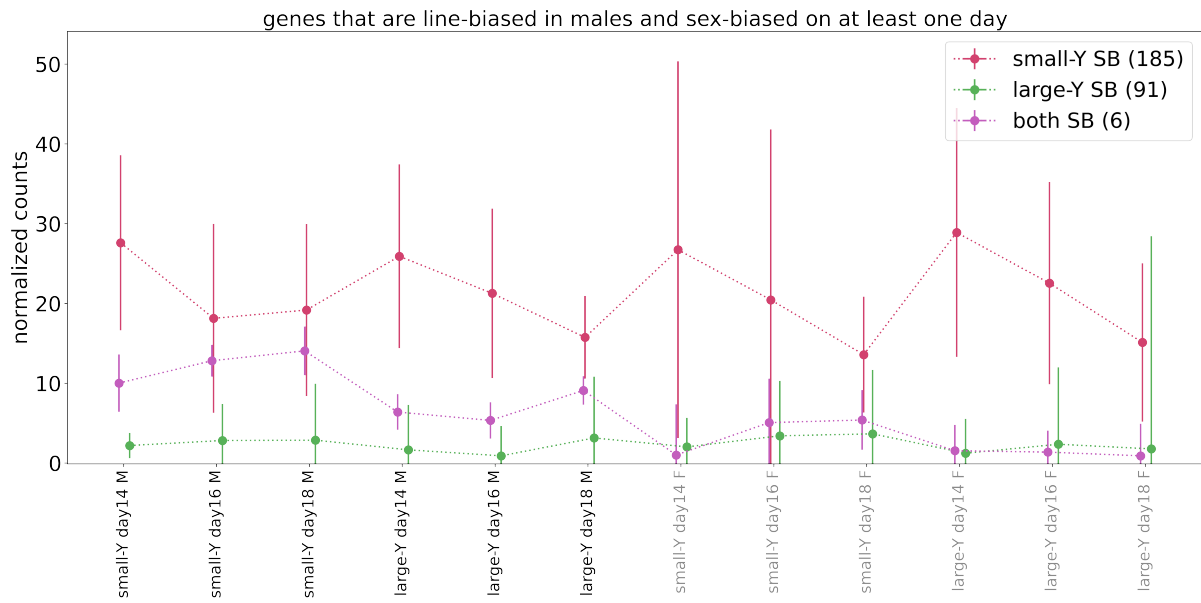


Figure S16: **Time-series plot of normalized counts of genes that are sex-biased and line-biased.** Genes that are significantly line-biased in males and sex-biased in at least one line (small-Y and large-Y sex bias shown as separate lines) on at least one day. Biological replicates and genes in the same significance intersection are merged, showing median and standard error in all conditions.



Figure S17: **GO-term enrichment and semantic clustering of genes that are significant for sex-bias in large-Y lines and line-bias in males on day 14.** The background of this is genes that are significant in large-Y sex bias, male line-bias or the sex-by-line interaction on day 14 (all genes shown in Fig. 4B).

Gene ID	Name	Description
gene-149137	-	no_description
gene-372264	-	—not_annotated—
gene-227137	-	no_description
gene-224956	RpL37A	Ribosomal L37ae protein family
gene-414353	-	no_description
gene-39545	Osi20	Protein of unknown function (DUF1676)
gene-115312	Tsf1	Transferrins are iron binding transport proteins which bind Fe(3) ion in association with the binding of an anion, usually bicarbonate
gene-130081	-	no_description
gene-371957	-	serine-type endopeptidase activity. It is involved in the biological process described with proteolysis
gene-204669	-	Lipase
gene-166391	-	DM4/DM12 family
gene-246615	-	—not_annotated—
gene-277078	GPRCAL3	G-protein coupled receptor activity. It is involved in the biological process described with G-protein coupled receptor signaling pathway
gene-406603	-	Ciliary basal body-associated, B9 protein
gene-158197	-	—not_annotated—
gene-122220	-	Belongs to the short-chain dehydrogenases reductases (SDR) family
gene-100036	-	Zinc-dependent metalloprotease
gene-143368	-	Zinc carboxypeptidase
gene-130096	-	no_description
gene-398993	-	Src homology 3 domains
gene-220249	-	axonemal central apparatus assembly

Table S1: Gene IDs and annotated functions (from eggnog) of genes sig. in both (blue) in Fig. 4B.



Figure S18: **GO-term enrichment and semantic clustering of genes that are significant for sex-bias in small-Y lines and line-bias in males on day 16.** The background of this is genes that are significant in small-Y sex bias, male line-bias or the sex-by-line interaction on day 16 (all genes shown in Fig. 4C).

Gene ID	Name	Description
gene-372264	-	—not_annotated—
gene-23884	-	—not_annotated—
gene-24278	-	—not_annotated—
gene-90918	-	—not_annotated—
gene-279676	-	Collagen triple helix repeat (20 copies)
gene-240602	-	no_description
gene-225158	-	—not_annotated—
gene-80062	-	—not_annotated—
gene-222332	-	Endoribonuclease L-PSP
gene-30595	-	—not_annotated—
gene-411056	-	Sarcolemmal associated protein
gene-329410	Grd	Belongs to the ligand-gated ion channel (TC 1.A.9) family
gene-431362	mRpL22	Ribosomal protein L22p/L17e
gene-13404	-	Haloacid dehalogenase-like hydrolase
gene-48598	-	glutathione s-transferase
gene-104371	-	no_description
gene-351334	-	cytoplasmic mRNA processing body assembly
gene-55869	-	Catalyzes the hydrolysis of N-formyl-L-kynurenine to L- kynurenine, the second step in the kynurenine pathway of tryptophan degradation. Required for elimination of toxic metabolites
gene-390687	-	—not_annotated—
gene-127707	-	UPF0193 protein
gene-63245	-	no_description
gene-240860	-	Ring finger domain
gene-23834	-	—not_annotated—
gene-122692	-	Phosphatidylcholine 1-acylhydrolase activity. It is involved in the biological process described with lipid metabolic process

Table S2: Gene IDs and annotated functions (from egglog) of genes sig. in both (blue) in Fig. 4C.

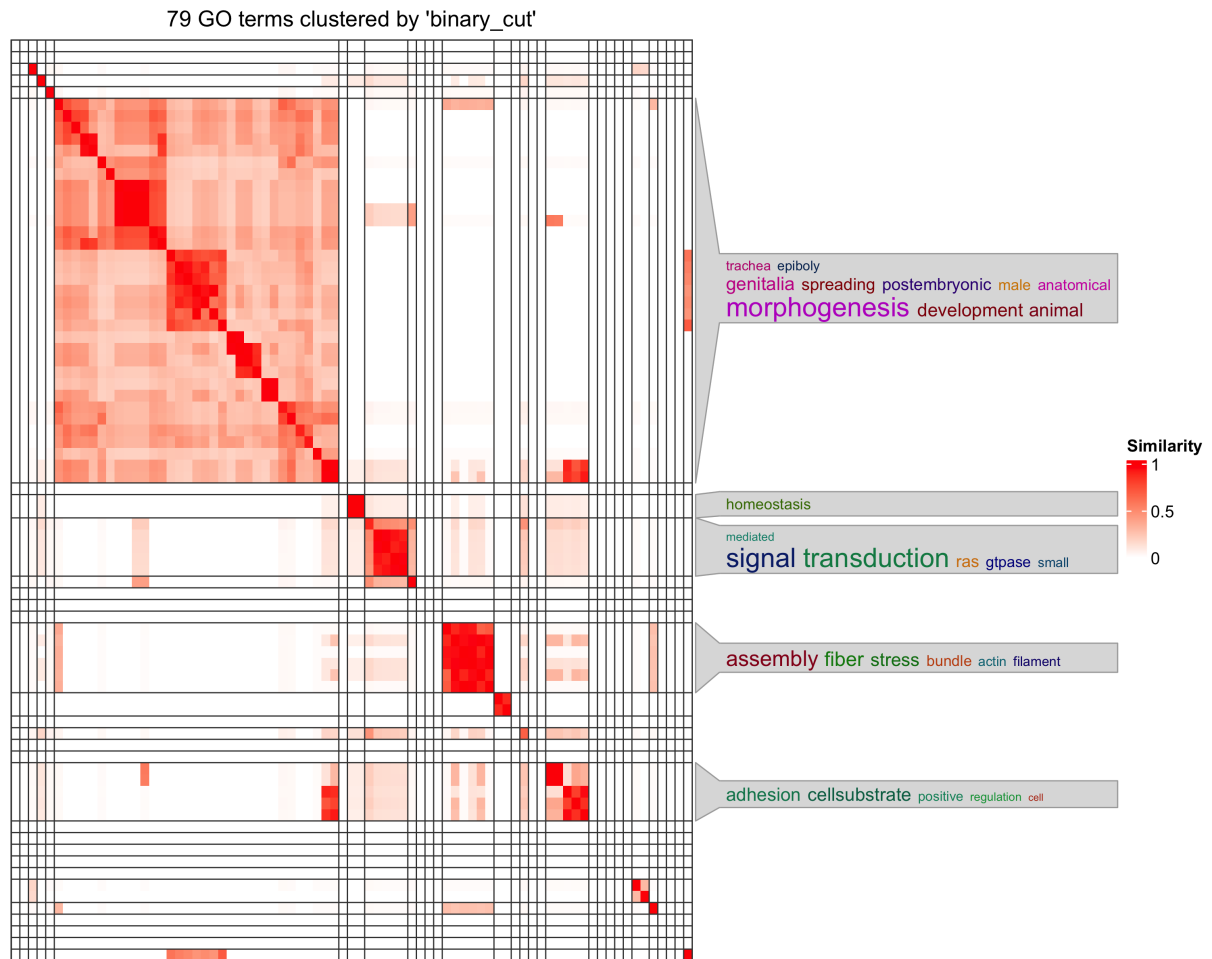


Figure S19: **GO-term enrichment and semantic clustering of genes that are significant for sex-bias in large-Y lines and line-bias in males on day 16.** The background of this is genes that are significant in large-Y sex bias, male line-bias or the sex-by-line interaction on day 16 (all genes shown in Fig. 4D).

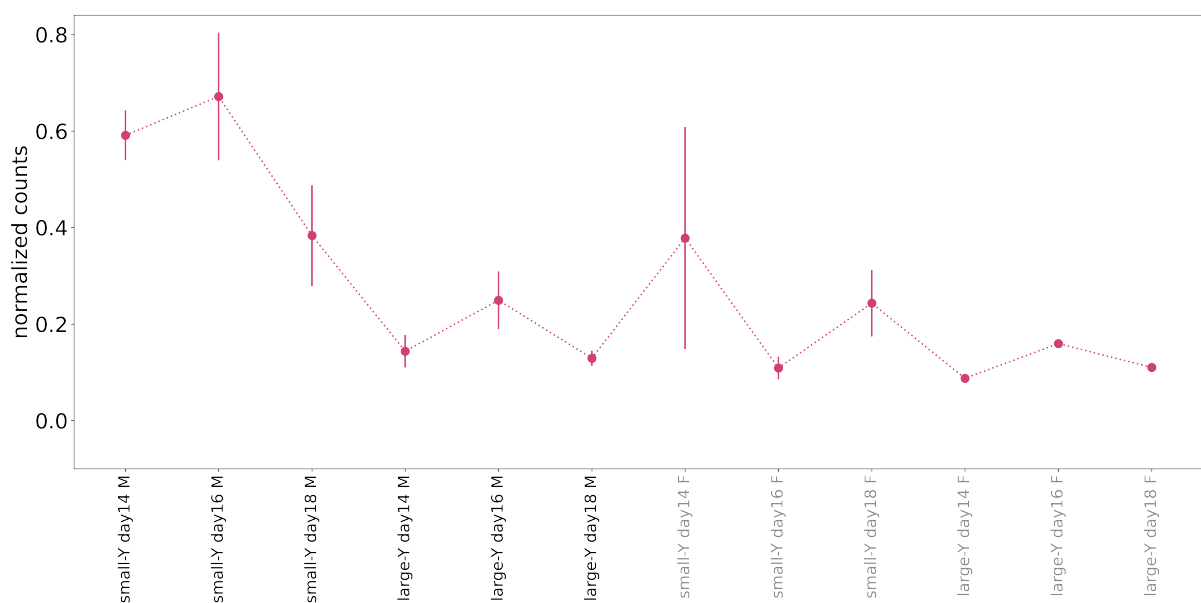


Figure S20: **y-TOR expression.**

Gene ID	Name	Description
gene-372264	-	—not_annotated—
gene-23413	Uba1	Belongs to the ubiquitin-activating E1 family
gene-24120	MED22	RNA polymerase II transcription cofactor activity. It is involved in the biological process described with regulation of transcription from RNA polymerase II promoter
gene-224357	-	Peptidase S24-like
gene-15763	-	7 transmembrane receptor (rhodopsin family)
gene-23365	Myo1B	Belongs to the TRAFAC class myosin-kinesin ATPase superfamily. Myosin family
gene-349163	-	Reverse transcriptase (RNA-dependent DNA polymerase)
gene-24203	-	Leucine-rich repeat-containing protein
gene-24221	-	C-terminal domain on Strawberry notch homologue
gene-24167	-	Transport and Golgi organisation 2
gene-23689	-	Formin Homology 2 Domain
gene-225030	Hsc70-5	Belongs to the heat shock protein 70 family
gene-12075	-	—not_annotated—
gene-24290	-	Conserved hypothetical ATP binding protein
gene-410366	-	no_description
gene-24052	Scm	Pfam:DUF3588
gene-423321	-	—not_annotated—
gene-23514	Uba1	Belongs to the ubiquitin-activating E1 family
gene-24079	-	—not_annotated—
gene-24088	Mob3	Mob1/phocein family
gene-7220	-	—not_annotated—
gene-24132	-	Guanylate cyclase
gene-23840	-	FKBP-type peptidyl-prolyl cis-trans isomerase
gene-23597	Prm	Myosin tail
gene-24185	-	Phosphotransferase activity, for other substituted phosphate groups. It is involved in the biological process described with phospholipid biosynthetic process
gene-23538	-	twin BRCT domain
gene-27466	-	GTP binding. It is involved in the biological process described with
gene-220249	-	axonemal central apparatus assembly

Table S3: Gene IDs and annotated functions (from egglog) of genes sig. in both (blue) in Fig. 4D.

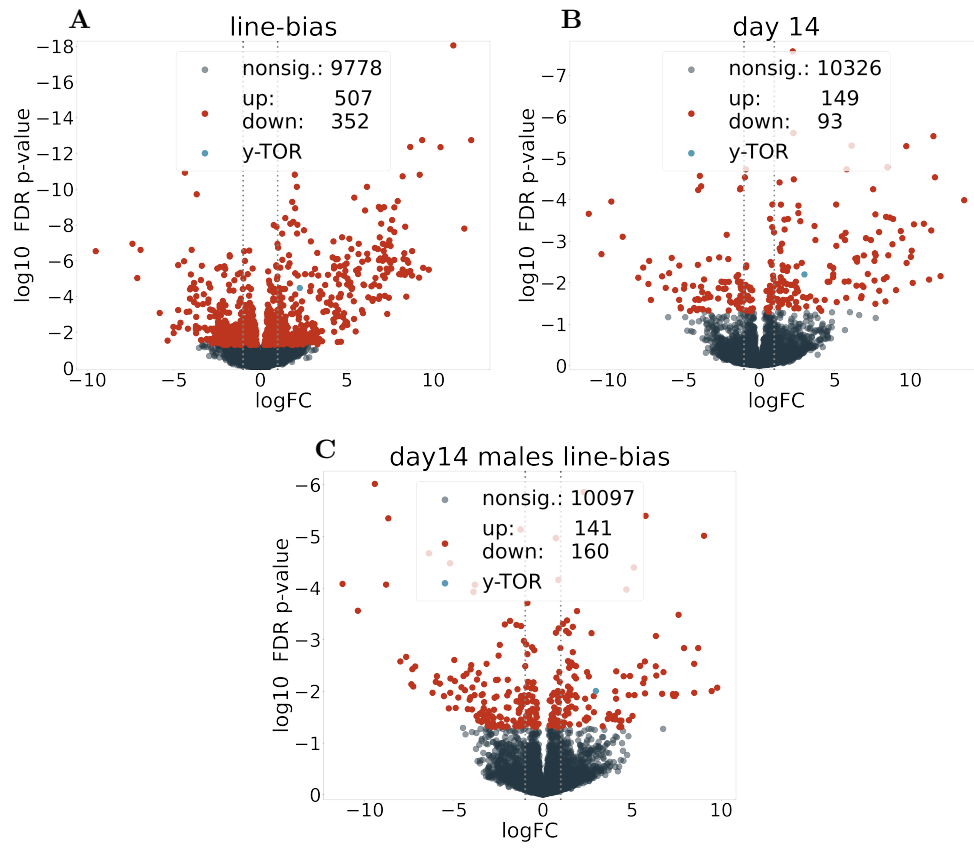


Figure S21: **Volcano plot of contrasts where y-TOR is significant.** (A): Using only male samples, specifying the days as coefficients and the contrast as small-Y - large-Y. (B): Using only male samples, specifying the day 14 small-Y - day14 large-Y contrast. (C): Using only day 14 samples, using the males small-Y - males large-Y contrast.